



The Analysis of Australian's International Trade

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Table of Contents

1. Introduction & Data Overview	1
2. National Trade Overview (1988–2024)	1
3. Trade Partners Overview	2
4. Trade Trends by Category (Export and Import)	3
4.1 Export and Import Trend Analysis in AUD	3
4.2 Export and Import Trend Analysis in %YoY Change	4
4.3 Export and Import Trend Analysis in % Proportion.....	5
5. Main Category Overview: Crude Materials (SITC2)	5
6. Dashboard: Subcategory Focus: Metalliferous Ores & Scrap (SITC 28)	6
7. Storyboard:	8
7.1 Slide 1: Australia’s Global Edge in Crude Material Exports	8
7.2 Slide 2: Which Subcategories Drive Crude Material Exports?	8
7.3 Slide 3: Metalliferous Ores: Winners and Laggards Across 36 Years.....	9
7.4 Slide 4: Volatility vs Stability in Export Growth – Subcategory 28 vs Others	9
7.5 Slide 5: How Global Events and Turning Points Reshaped Australia’s Ore Export.....	10
7.6 Slide 6: Iron Ore: Australia’s Star Performer (Subcategory 28)	10
7.7 Slide 7: Future Outlook for Metalliferous Ores	11
7.8 Slide 8: Why Iron Ore Should Remain a National Priority.....	11
7.9 Slide 9: Is Australia a Net Exporter in Crude Materials?	12
7.10 Slide 10: What Does Australia Import Under Crude Materials?	12
7.11 Slide 11: Balancing Export Strength with Import Needs	13
8. Deep-Dive Product Focus: Iron Ore Exports.....	14
8.1 5-Year Forecast (2025–2030).....	15
8.2 Recommendations	15
9. Visualization Approach Summary	16
10. Advantages and Disadvantages of the Dashboard and Storyboard Methodologies.....	16
11. Summary and Conclusion	17
12. Reference	18

1. Introduction & Data Overview

This report investigates Australia’s international trade from 1988 to 2024 using the Australian Bureau of Statistics (ABS) data. The dataset includes export and import values across ten main product categories and sixty-seven subcategories based on the Standard International Trade Classification (SITC). The analysis focuses on long-term trade trends and structural changes, covering key global events such as the Asian Financial Crisis, the Global Financial Crisis, and the COVID-19 pandemic. The period allows for a comprehensive view of economic shifts that have influenced Australia’s trade position.

2. National Trade Overview (1988–2024)

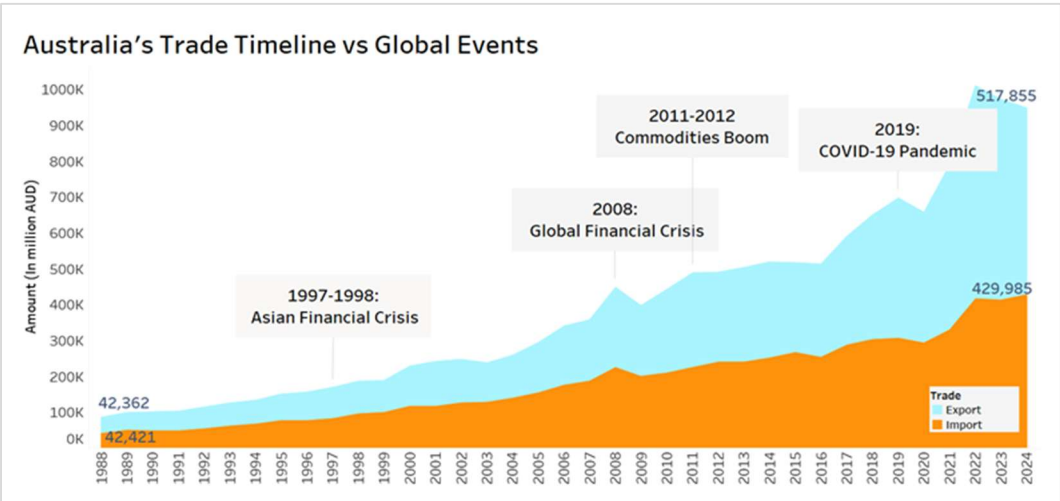


Figure 2.1 Stacked Area, Australia’s Export and Import Trends (1988–2024)

Australia’s international trade has grown significantly over the past three decades. The first visual (*Figure 2.1*) shows export and import values from 1988 to 2024. In 1988, export and import values were nearly equal, with exports at AUD 42,421 million and imports at AUD 42,362 million. However, the gap widened over time, especially after 2010, as exports began to outpace imports, leading to trade surpluses. This growth accelerated after the 2019 COVID-19 pandemic, with exports reaching AUD 517,855 million in 2024 and imports at AUD 429,985 million.

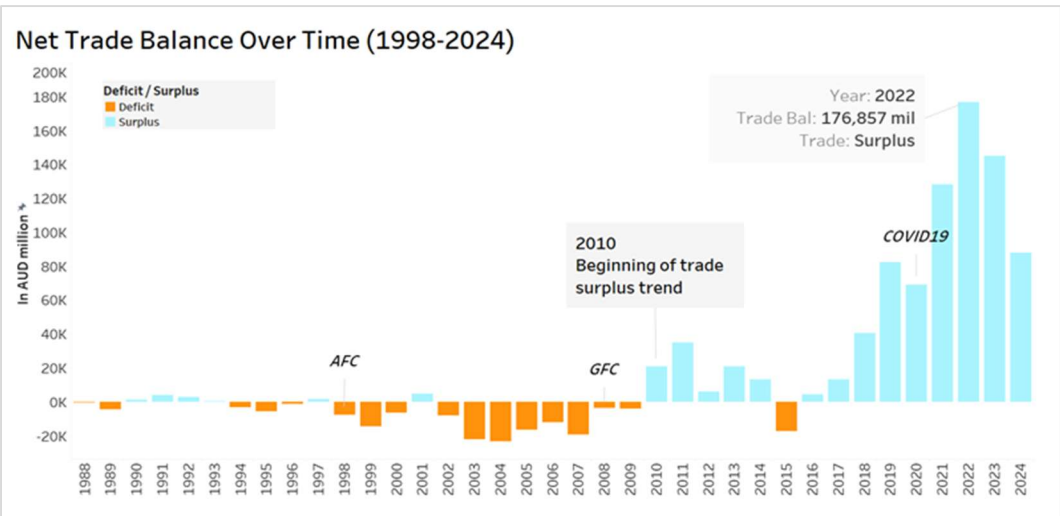


Figure 2.2 Trade Balance Over Time (1988–2024)

In *Figure 2.2*, we focus on the net trade balance. From 1988 to the mid-2000s, Australia experienced frequent trade deficits, as shown in the orange bars. This changed around 2010 when a consistent trade surplus trend began. This shift aligns with the post-GFC recovery and rising global demand for Australian commodities. The highest surplus occurred in 2022, at AUD 176,857 million. Some years, like 2015 and the early 2000s, still recorded minor deficits, but the overall trend moved strongly towards surplus after 2010, as mentioned in the previous graph

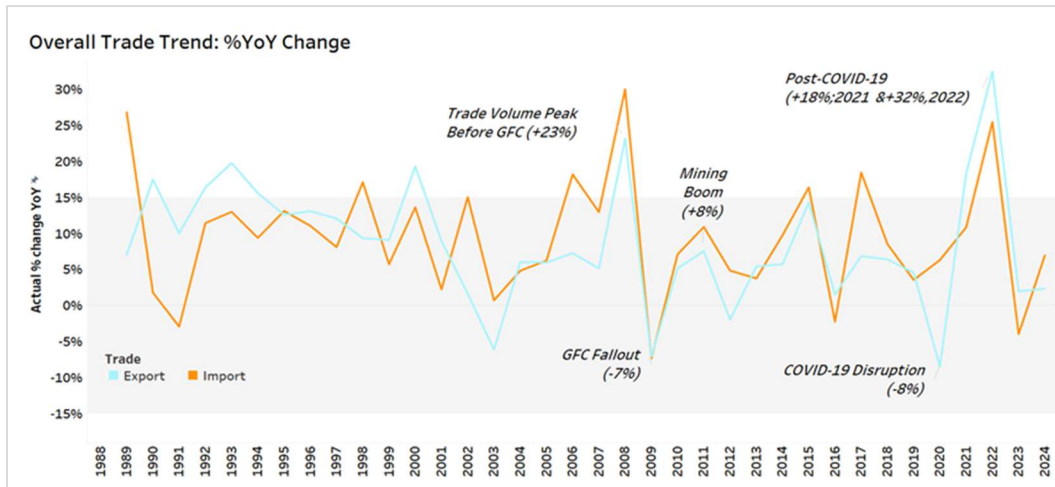


Figure 2.3 Trade Balance Over Time (1988–2024)

In *Figure 2.3*, a dual-line graph, presents year-over-year percentage changes in exports and imports. The data show several sharp movements, with a 23% trade volume peak before the Global Financial Crisis (GFC) and a 7% fall during the GFC period. After 2010, moderate growth was observed during the mining boom (+8%). However, exports and imports dropped by around 8% during the COVID-19 disruption. A strong rebound followed, with post-pandemic growth recorded at +18% in 2021 and +32% in 2022. This recovery reflects renewed global demand and improved supply chain stability (ABS, 2022).

These graphs provide a clear overview of how Australia's trade evolved in response to major global events. The steady rise in exports, particularly after 2010, highlights the country's strong position as a key exporter of raw materials. Meanwhile, the shift from deficit to surplus in the trade balance suggests an improved national trade performance over time.

3. Trade Partners Overview



Figure 3.1: Australia's main Export partner

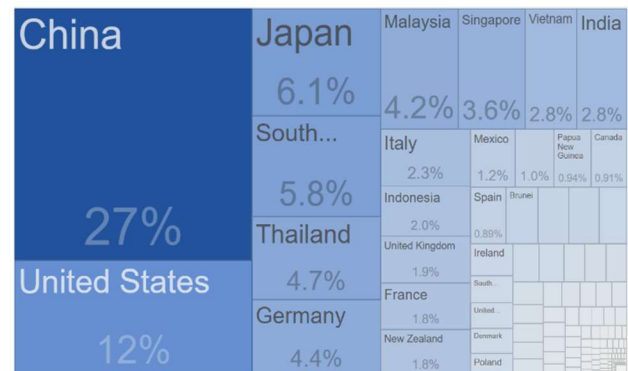


Figure 3.2: Australia's main import partners

(source: [Australia Imports By Country](#))

Australia's international trade is highly concentrated in a few major countries, with Asia playing a central role in exports and imports.

The first treemap (*Figure 3.1*) shows Australia's export destinations in percentage share. **China** dominates with 37% of total exports, followed by **Japan** (11%) and **South Korea** (7.2%) (Trading Economics, n.d.a). These top three countries alone account for over half Australia's export value. This heavy reliance on Asian markets reflects Australia's role as a key supplier of raw materials, especially in countries with large manufacturing industries. The strong relationship with China also helps explain the significant trade surpluses shown in *Figure 2.2*, particularly after 2010.

The second treemap (*Figure 3.1*) displays Australia's import partners. **China** again holds the largest share at 27%, followed by the **United States** (12%) and **Japan** (6.1%). Interestingly, the list of import partners is more diversified, with contributions from European countries such as **Germany** (4.4%), **France** (1.8%), and **Italy** (2.3%), as well as ASEAN members like **Malaysia** and **Singapore** (Trading Economics, n.d.b), which reflects Australia's import dependence on advanced economies and regional production hubs for machinery, electronics, and consumer goods.

The strong trade ties with **China** on both sides also explain the country's sensitivity to geopolitical shifts or supply chain disruptions related to **China**, raising strategic considerations around diversification.

4. Trade Trends by Category (Export and Import)

4.1 Export and Import Trend Analysis in AUD

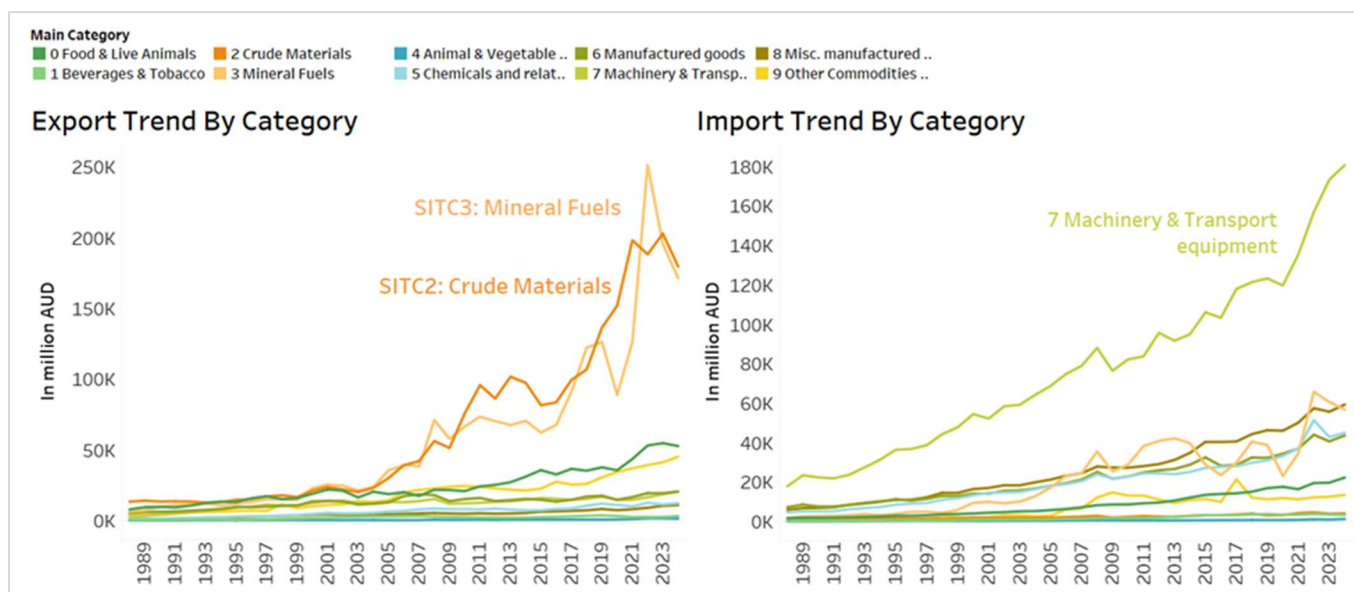


Figure 4.1: Australia's Export and Import Trends in AUD

Figure 4.1 illustrates Australia's long-term trade performance by category between 1989 and 2023. On the export side, **Mineral Fuels (SITC 3)** and **Crude Materials (SITC 2)** emerge as the leading contributors. Their values rose significantly after 2000 and accelerated further post-2010, primarily driven by rising demand from China and the impact of the mining boom. Mineral Fuels peaked around 2022, coinciding with global energy instability following the COVID-19 crisis. In contrast, other categories, such as Food and Manufactured Goods, showed slower and more modest growth.

On the import side, **Machinery and Transport Equipment (SITC 7)** has consistently remained dominant since the early 1990s, with sharp growth observed after 2010. This trend underscores Australia's persistent reliance on foreign technology and industrial products (ABS, 2023). While imports in other categories have also increased, none have approached the scale of Category 7.

Overall, the contrasting trajectories between exports and imports reinforce Australia's position as a major exporter of natural resources and a significant importer of high-value manufactured goods. This pattern aligns with the trade balance shifts and partner relationships discussed in previous sections.

4.2 Export and Import Trend Analysis in %YoY Change

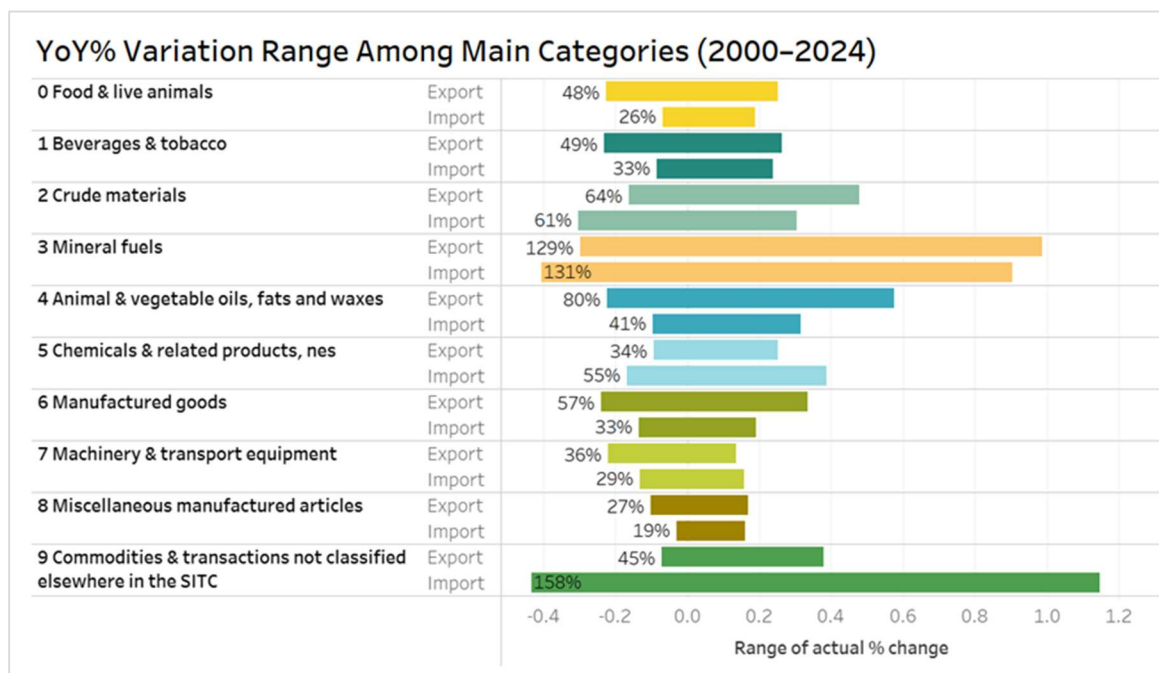


Figure 4.2: Australia's percentage range of YoY changes in exports and imports

Figure 4.2 illustrates the year-over-year percentage variation in exports and imports across main trade categories from 2000 to 2024. The analysis identifies categories with high volatility, which are critical for understanding trade-related risks and strategic planning.

Mineral Fuels (SITC 3) display the highest variation, 129% for exports and 131% for imports, highlighting the category's sensitivity to global energy markets, geopolitical instability, and post-COVID disruptions, which reinforces its high-risk, high-reward profile, as previously seen in *Figure 2.3*.

Crude Materials (SITC 2) also show considerable fluctuation (64% export, 61% import), primarily driven by Subcategory 28, a major export contributor. These figures reflect exposure to commodity pricing and demand shifts, especially from China.

Animal and vegetable Oils (SITC 4) reveal 80% export variation, suggesting links to seasonal output and global food markets. On the import side, **Unclassified Commodities** (SITC 9) show extreme volatility (158%), likely due to irregular, low-volume transactions.

In contrast, **Manufactured Goods** (SITC 6) and **Machinery and transport Equipment** (SITC 7) exhibit more stable patterns, consistent with Australia's role as a major importer of industrial products. Overall,

the most volatile categories are also high-value exports, underscoring the need to balance opportunity with exposure.

4.3 Export and Import Trend Analysis in % Proportion

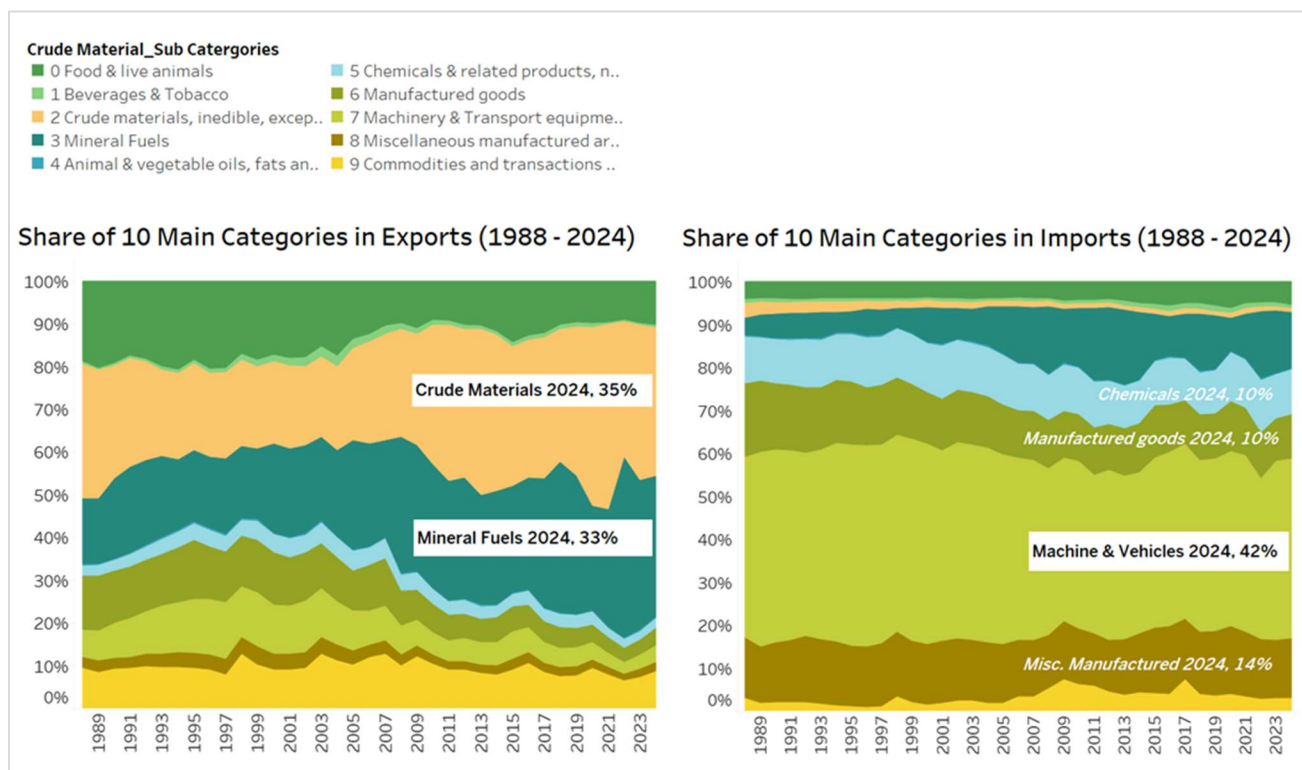


Figure 4.3: Area charts showing proportion of Main categories in both export and import

The proportion charts in *Figure 4.3* provide a structural view of Australia's trade composition from 1988 to 2024. On the export side, **Crude Materials** and **Mineral Fuels** have maintained the most significant shares, making up nearly 70% of total exports in 2024. This reinforces previous findings that Australia's export strength is highly concentrated in resource-based sectors.

In contrast, Machinery and transport Equipment dominate the import structure, accounting for 42% of total imports in 2024. This stable dominance highlights Australia's long-term reliance on importing industrial and technological goods, consistent with the partner trends discussed earlier in *Figure 3.2*.

5. Main Category Overview: Crude Materials (SITC2)

Crude Materials (SITC 2) have shown steady growth over the past two decades, as seen in the national trade overview (*Figure 4.1*). In 2024, they accounted for 35% of Australia's total exports, driven mainly by demand from Asian markets. This category generates substantial export revenue and supports Australia's trade surplus.

To understand what drives this performance, the following section closely examines **Subcategory 28: Metalliferous Ores & Scrap**, which plays a leading role within SITC 2.

6. Dashboard: Subcategory Focus: Metalliferous Ores & Scrap (SITC 28)

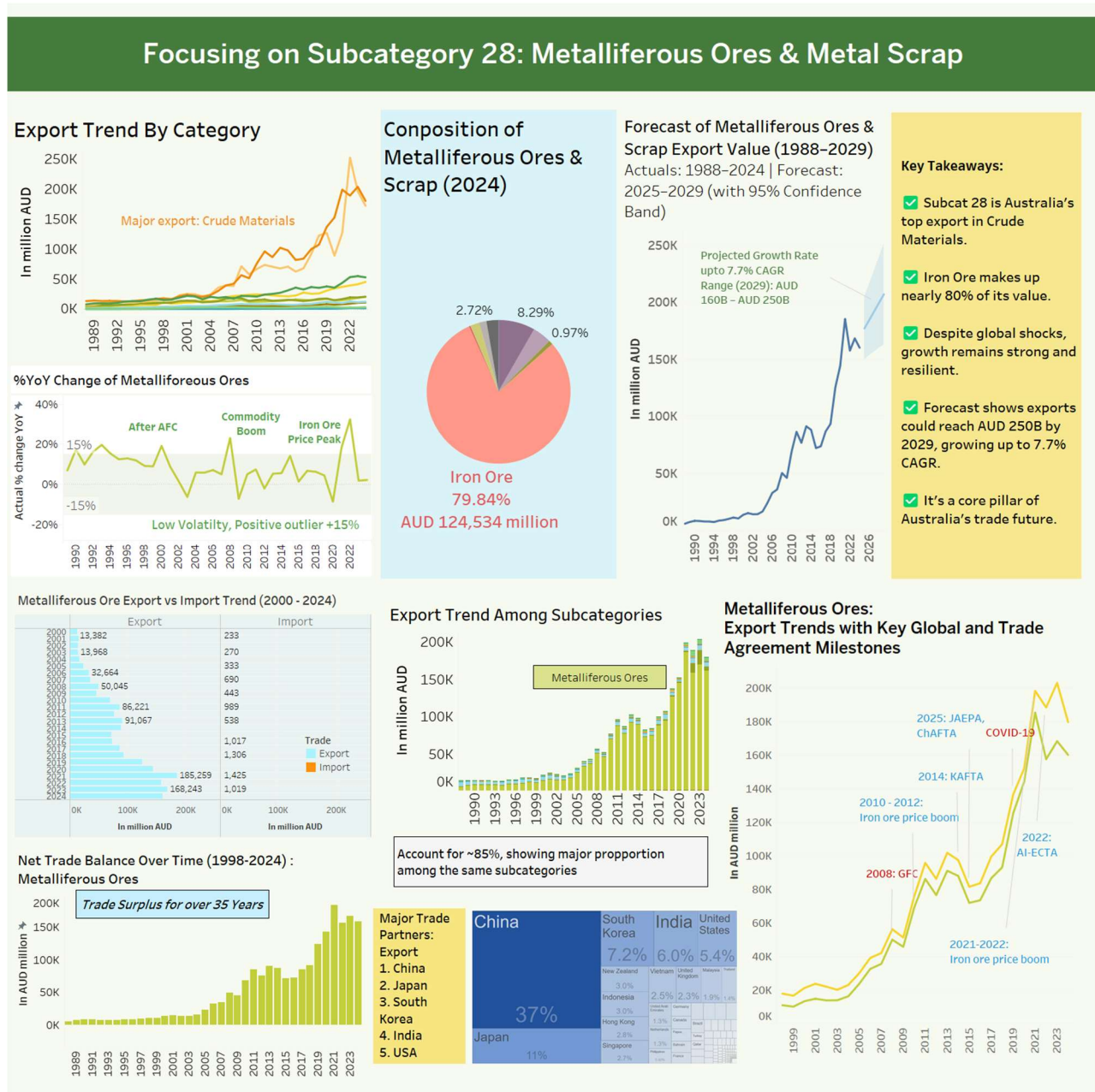


Figure 6: Dashboard overview of Subcategory28: Metalliferous Ores

According to this dashboard (Figure 6), each visual was crafted to support different types of analysis, helping users uncover long-term performance patterns, structural characteristics, and future implications within this critical sector.

Comparative Line Chart: Framing the Sector's Dominance

I began with a line chart comparing export values across main categories, which helped position **Crude Materials**, and by extension, **Subcategory 28**, as a leading contributor in Australia's export profile, providing a contextual anchor for deeper analysis.

Composition Pie Chart: Revealing Internal Structure

To clarify the internal composition of **Subcategory 28**, I created a pie chart based on 2024 export values. It immediately reveals **Iron Ore**'s overwhelming share of nearly 80%, which helped guide my decision to make it the primary focus of follow-up analyses.

Export vs Import Bar Chart: Highlighting Net Value Advantage

To provide a more precise comparison between trade flows, I added a side-by-side bar chart displaying export and import values of Subcategory 28 from 2000 to 2024. This tool offers a direct visual contrast, effectively reinforcing the subcategory's strong net export position throughout the period.

Net Trade Balance Chart: Showing Long-Term Economic Value

This vertical bar chart tracks the trade balance of **Metalliferous Ores** from 1998 onward. I used this to highlight the subcategory's contribution to over 35 years of uninterrupted trade surplus, especially after 2010. It serves as a compelling metric of economic resilience.

Stacked Bar Chart: Monitoring Stability Within the Subcategory

To dive deeper into subcomponent dynamics, I developed a stacked bar chart showing trends across the subcategories. It allowed me to visualise the consistency of metalliferous ores as a stable backbone, not just a high-growth outlier.

Event-Overlay Line Graph: Linking Data with Real-World Context

I designed a line graph annotated with economic events and trade agreements to connect trade data with a global context, which gave me a way to map how events like the GFC, COVID-19, and FTA milestones corresponded with observable shifts in export value.

YoY Change Chart: Identifying Volatility

This compact line chart of year-on-year percentage change helped me quickly assess volatility. Notably, I observed a +15% outlier during recovery periods, highlighting the cyclical nature of the market and potential demand surges.

Forecast Model: Exploring Forward Scenarios

I used exponential smoothing for forward-looking analysis to build a forecast chart through to 2029. A 95% confidence band allowed me to project export values ranging from AUD 190B to 300B while recognising underlying uncertainty in commodity markets.

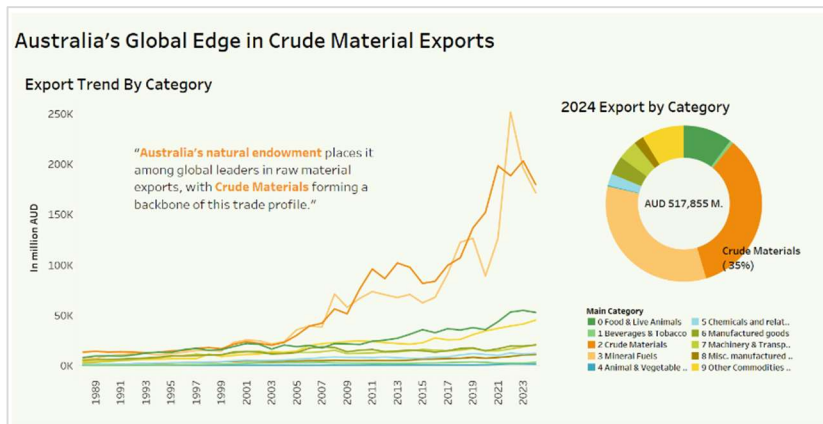
Strategic Summary Box: Synthesising Key Findings

Finally, I added a summary box to distil the dashboard's output into high-level takeaways. While not a visual tool per se, it serves as a quick reference point for stakeholders, combining analytical depth with strategic clarity.

Overall, I designed this dashboard as a storytelling medium and an interactive analytical platform. By combining diverse tools, from composition and forecasting to volatility and event linkage, I aimed to give users the freedom to explore, interpret, and evaluate this subcategory from multiple angles. Doing so empowers more informed discussions on the future of Australia's trade performance in metalliferous ores.

7. Storyboard:

7.1 Slide 1: Australia's Global Edge in Crude Material Exports



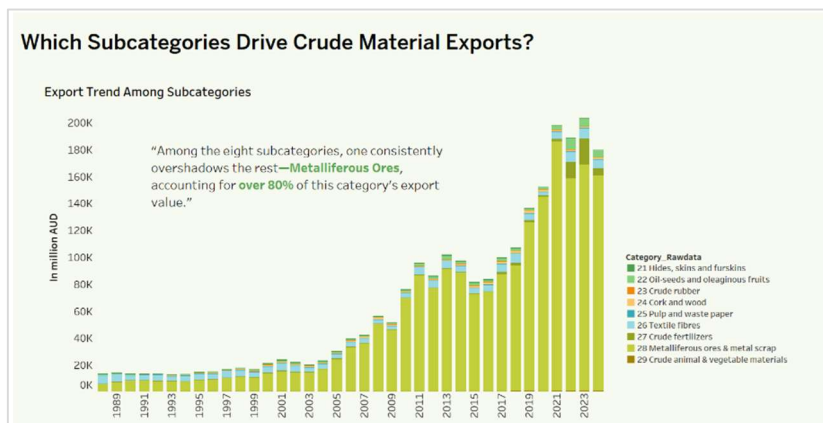
This introductory slide was designed to provide a high-level overview of Australia's export composition over time. I first created a line chart to visualise raw dollar export trends from 1988 to 2024 across all main categories. Then, I complemented this with a donut chart showing the export share of each category in 2024. The aim was to immediately establish Crude Materials's

prominence in historical and current contexts.

A multi-line chart was used to track long-term export performance by category, the use of consistent colour coding allowed for clear category distinction. The donut chart provided a snapshot of the 2024 export distribution, effectively highlighting that Crude Materials contributed 35% of total export value, making it the top-performing category. Labeling and interactive tooltips enhanced interpretability.

The visualisation clearly shows that Crude Materials have seen the most substantial and sustained growth since the early 2000s, significantly outpacing other categories. This upward trend aligns with Australia's natural resource advantage and high global demand, especially from Asia. In 2024, Crude Materials accounted for over a third of Australia's export value, underscoring their economic significance and justifying their role as the analytical focus for the rest of the storyboard.

7.2 Slide 2: Which Subcategories Drive Crude Material Exports?



I created a stacked bar chart using export data for all eight subcategories from 1988 to 2024 to identify the subcategories contributing most to Crude Material exports. The purpose was to illustrate both the overall growth of the category and the dominant role of Subcategory 28 (Metalliferous Ores) in that growth.

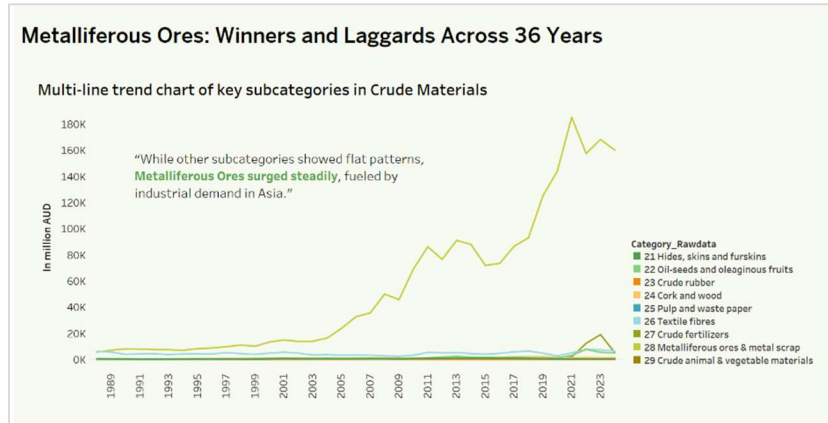
A stacked vertical bar chart was chosen to visualise the cumulative

value of all subcategories over time. Each stack segment was colour-coded by subcategory, allowing for clear visual comparison. This method made it easy to observe which subcategories consistently contributed most to total exports, especially in recent years.

The chart shows that Metalliferous Ores (Subcategory 28) has overwhelmingly driven the growth of Crude Material exports. Since the early 2000s, this subcategory has expanded rapidly, and by 2024, it accounts

for over 80% of the category's total export value. Other subcategories, such as hides, oil seeds, and pulp materials, contribute much smaller proportions, highlighting a firm reliance on a single export segment. This dominance sets the stage for further investigation into Subcategory 28 throughout the storyboard.

7.3 Slide 3: Metalliferous Ores: Winners and Laggards Across 36 Years



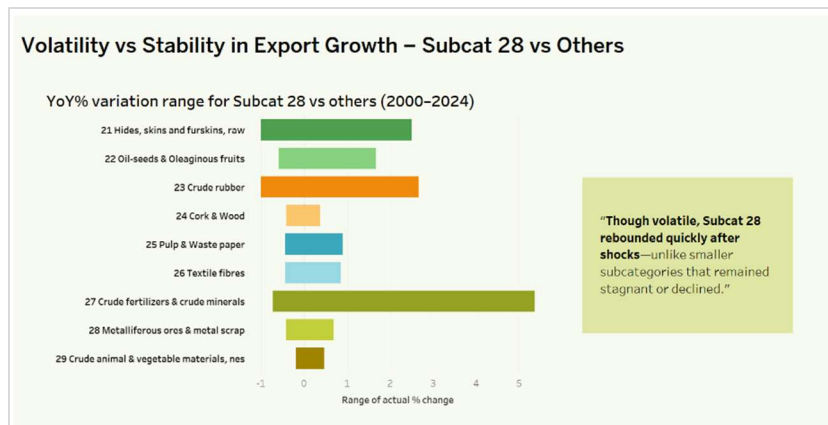
This slide was created to compare long-term export performance across all subcategories within Crude Materials. By plotting each subcategory as a separate line over the period 1988 - 2024, I aimed to visualise overall growth and disparities in subcategory performance.

A multi-line trend chart was used to plot export values for all eight subcategories over 36 years. Each

line represents a subcategory, colour-coded consistently with previous slides. This technique distinguishes between “**leaders**” and “**laggards**” over time, with Metalliferous Ores visually standing out in both trajectory and scale.

The chart reveals that while most subcategories exhibit flat or modest growth trends, Metalliferous Ores (Subcategory 28) shows an extraordinary upward trajectory, especially from 2005 onward, indicating a breakthrough shift in export volume and demand. This growth pattern reflects increased industrial activity in Asia, particularly from China, and further reinforces why this subcategory has become the cornerstone of Australia's raw material trade.

7.4 Slide 4: Volatility vs Stability in Export Growth – Subcategory 28 vs Others



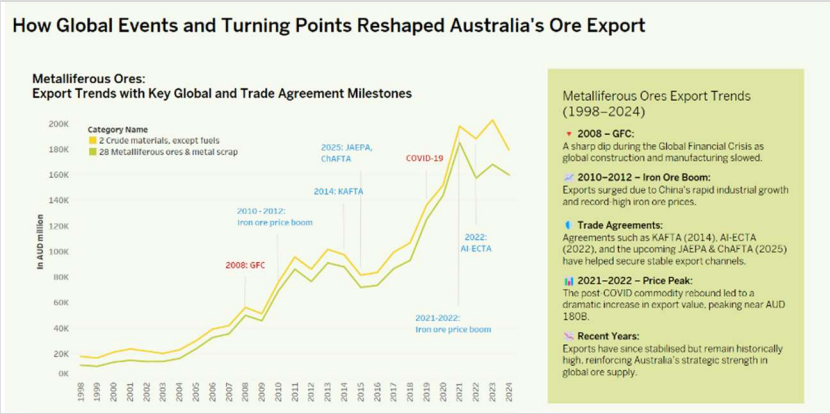
To examine the degree of export fluctuation across subcategories, I calculated the range of Year-on-Year percentage change from 2000 to 2024. I then created a horizontal bar chart to compare the volatility of each subcategory over time, which helped reveal which segments were most stable or unstable in their export growth patterns.

A horizontal bar chart visualised the range between maximum and minimum %YoY change for each subcategory. This format allowed for an intuitive left-to-right comparison of volatility. The bars were sorted from most to least volatile, with consistent colour mapping to previously used category tones.

The chart shows that while some subcategories, like Crude fertilisers (27) and Crude rubber (23), experienced wide fluctuations, Subcategory 28 (Metalliferous Ores) had moderate volatility but

consistently rebounded after major shocks. This pattern indicates both exposure to global market forces and resilience. In contrast, several smaller subcategories remained relatively flat, reflecting limited trade momentum or stable but stagnant performance.

7.5 Slide 5: How Global Events and Turning Points Reshaped Australia’s Ore Export



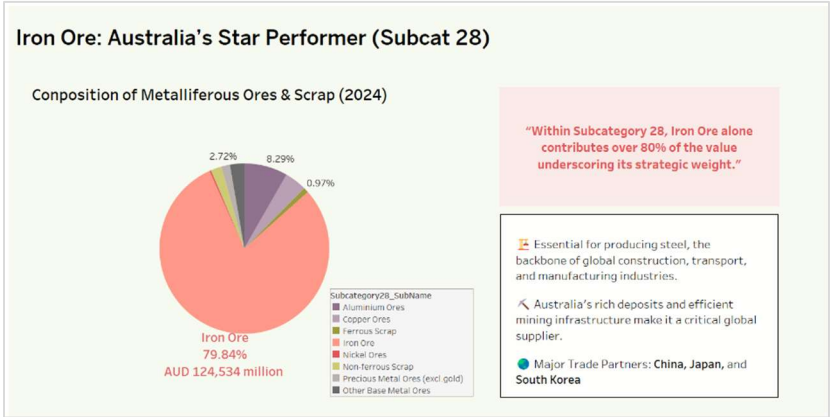
This slide was created to link global events and trade policy milestones to shifts in export performance, focusing on Subcategory 28 and its parent category. I annotated key years on a dual-line chart that compares export values of Crude Materials (Category 2) and Metalliferous Ores (Subcategory 28) from 1998 to 2024. Each turning point, such as economic crises, price booms, and trade

agreements—was integrated directly into the timeline.

A dual-line trend chart was used to show category and subcategory export trends in the same visual space. I overlaid milestone annotations using coloured text and icons to distinguish between types of events (e.g., GFC, COVID-19, trade deals). This combination of trend and annotation transforms the chart into a narrative timeline that connects data patterns to real-world causes.

The visual shows that export dips and surges closely align with major global or regional events. For example, the 2008 GFC caused a sharp drop, while the 2010-2012 iron ore boom and post-COVID rebound in 2021-2022 triggered export spikes. Introducing key trade agreements like KAFTA, AI-ECTA, and the upcoming JAEPA/ChAFTA (DFAT, 2023) appear to align with sustained export growth, supporting the idea that policy stability reinforces market resilience, which reinforces the strategic role of Subcategory 28 as both economically responsive and policy-sensitive.

7.6 Slide 6: Iron Ore: Australia’s Star Performer (Subcategory 28)

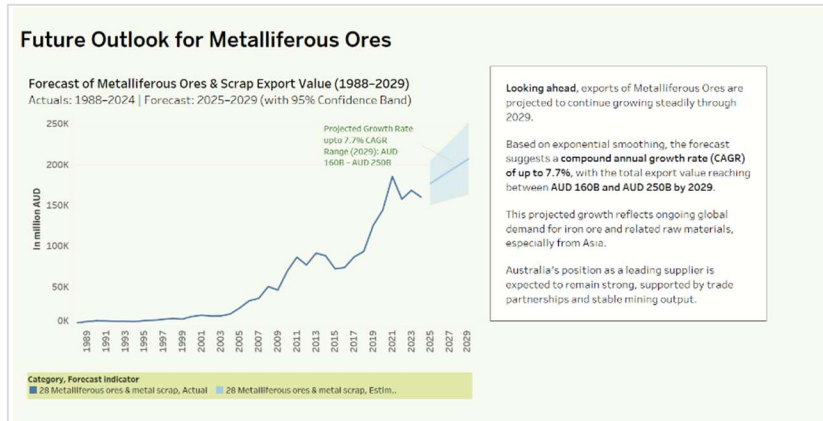


This slide focuses on identifying the key product within Subcategory 28. I used 2024 export data (ABS, 2024) to categorise the subcategory into its constituent product types. I visualised this using a pie chart to highlight the overwhelming contribution of iron ore compared to other ores and metal scrap.

A pie chart was chosen to display the relative share of each product in Subcategory 28. Iron Ore was visually emphasised using a bold colour and larger label, representing nearly 80% of the subcategory’s total export value. Supplementary text and icon-based highlights reinforce iron ore’s industrial importance and key export destinations.

The visual shows that Iron Ore is the undisputed driver of Subcategory 28, contributing over AUD 124 billion in export value in 2024 alone (ABS, 2024). Its dominance reflects Australia's natural advantage in iron ore reserves, mining infrastructure, and global reliance on steel production. The three major trade partners (China, Japan, and South Korea) also align with global industrial demand, reinforcing the geopolitical and economic significance of this product.

7.7 Slide 7: Future Outlook for Metalliferous Ores

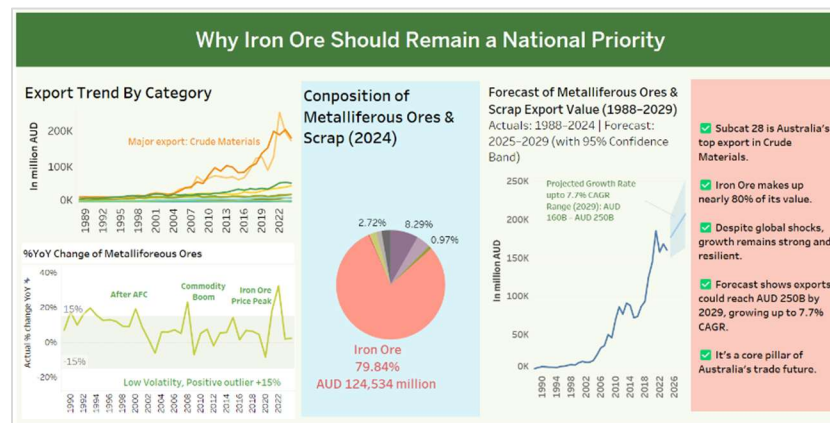


To project future export trends, I applied exponential smoothing to the historical export values of Subcategory 28 from 1988 to 2024, generating a forecast from 2025 to 2029. The chart includes a 95% confidence interval, giving a range of plausible export outcomes and helping to quantify uncertainty around the growth trend.

A time-series line chart was used with two layers: actual values (1988–2024) and forecasted values (2025–2029). A shaded confidence band highlights the upper and lower bounds of the forecast. At the same time, an explicit annotation indicates the expected Compound Annual Growth Rate (CAGR) of up to 7.7%, with a projected range of AUD 160B to AUD 250B by 2029.

The forecast suggests continued export growth, reinforcing Subcategory 28's strategic importance. The projected CAGR aligns with the ongoing global demand for iron ore, particularly from industrialised Asian economies. Australia's stable mining infrastructure and strong trade partnerships will likely support this trend, positioning the country to remain a dominant global supplier well into the future.

7.8 Slide 8: Why Iron Ore Should Remain a National Priority



This slide is a visual summary dashboard that integrates all key insights from the preceding storyboard. It combines export trends, product composition, year-on-year growth, and forecasted future value centred around Subcategory 28 and its key product, iron ore. The layout was carefully designed to show the whole narrative arc in one glance: from historical dominance to future projections.

projections.

A dashboard layout was used, featuring four key visualisations:

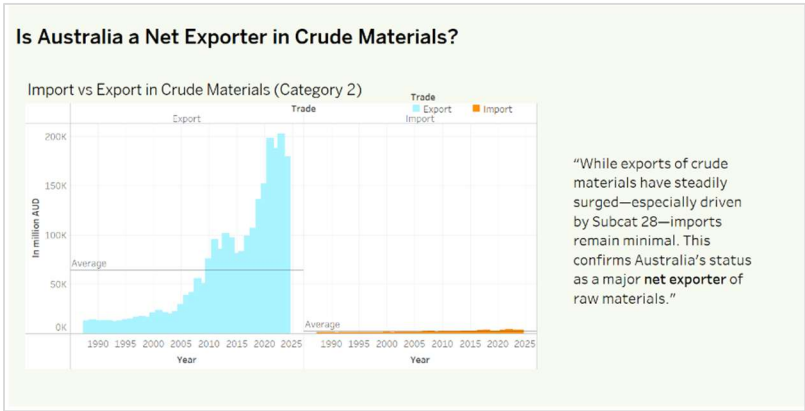
- A line chart to highlight Category 2's long-term export dominance
- A YoY change chart to show growth volatility

- A pie chart displaying iron ore’s product share in 2024
- A forecast line chart with 95% confidence bands

Text boxes and checklist-style annotations were added to reinforce major conclusions in a visual and digestible format.

This integrated view reinforces the central argument: Iron Ore is the engine of Australia's Crude Material exports. With nearly 80% share in Subcategory 28, strong post-shock recovery, and a projected CAGR of 7.7%, iron ore’s role as a trade pillar is undisputed. Despite global uncertainty, its outlook remains robust, and its export performance justifies continued strategic support and national prioritisation.

7.9 Slide 9: Is Australia a Net Exporter in Crude Materials?



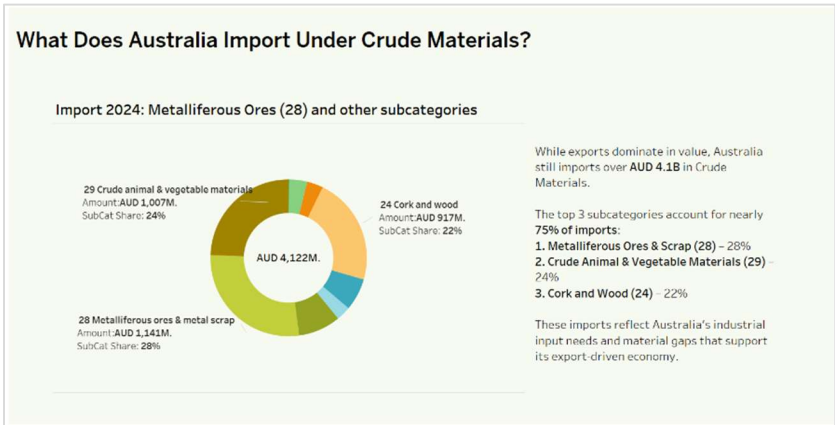
To assess Australia's trade balance in Crude Materials (Category 2), I created a split bar chart showing side-by-side annual values of exports and imports from 1988 to 2024. The goal was to evaluate whether Australia is a net exporter and how Subcategory 28 may have contributed to this position.

A dual-panel vertical bar chart was used to clarify exports and imports.

Light blue bars represented exports, while orange bars indicated imports. An average reference line was added for both panels to highlight the sustained gap. Colour differentiation and label positioning ensured the immediate interpretability of trade imbalances.

The visual clearly shows that exports of Crude Materials have grown substantially, particularly from the mid-2000s onward, while imports have remained negligible by comparison. This persistent gap confirms that Australia is a strong net exporter of raw materials, with Subcategory 28 playing a dominant role in generating the surplus. The chart reinforces Australia’s natural resource advantage and minimal dependence on imports in this category.

7.10 Slide 10: What Does Australia Import Under Crude Materials?



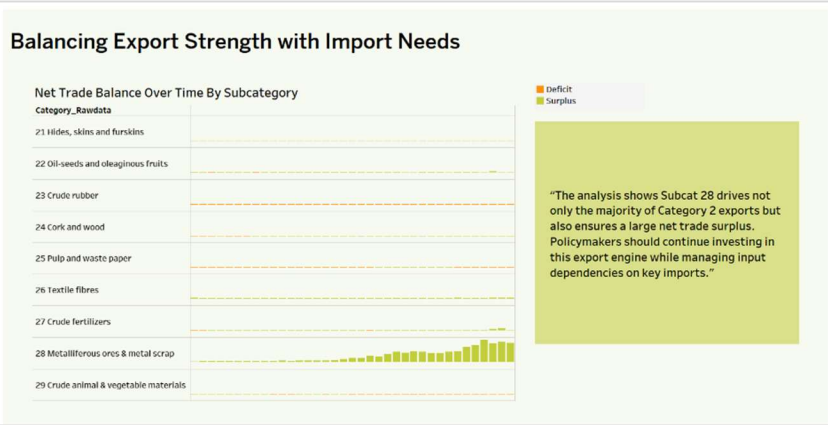
This slide shifts the lens to the import side of Crude Materials, focusing on identifying the most significant subcategories. Using a donut chart, I used 2024 data to visualise the total import value and its composition across subcategories, highlighting the top 3 contributors.

A donut chart represented the proportion of import values among subcategories. Values and

percentages were labelled directly on the chart for ease of comparison. The central label (AUD 4.1B) provides immediate context for the total import scale, while the outer segments clarify the most relevant subcategories.

Although exports far exceed imports, Australia still imported over AUD 4.1 billion in Crude Materials in 2024. The top three subcategories, Metalliferous Ores and scrap (28), Crude Animal and Vegetable Materials (29), and Cork and Wood (24), together made up over 75% of all imports. These figures reflect strategic input needs for Australia’s manufacturing and resource sectors, showing that even a major exporter must rely on targeted imports to support its broader trade infrastructure.

7.11 Slide 11: Balancing Export Strength with Import Needs



To summarise Australia’s trade efficiency across Crude Materials, I created a net trade balance chart by subcategory, calculating export value minus import value over time. The goal was to assess which subcategories yield surpluses and which contribute to trade deficits—particularly highlighting the role of Subcategory 28.

A diverging bar chart visualised trade surplus (in green) and deficit (in orange) across all subcategories. Each row represents a subcategory, with colour-coded bars indicating whether the trade balance was positive or negative across the years. This format offers a quick overview of economic contribution and dependency.

The analysis reveals that Subcategory 28 consistently delivers a substantial trade surplus, unlike other subcategories that remain in deficit. This underscores its role as a top export category and a stabilising force in Australia’s overall trade position. The chart also highlights the need to manage subcategories with persistent deficits strategically, reinforcing the importance of targeted import substitution or diversification policies.

8. Deep-Dive Product Focus: Iron Ore Exports

This section offers a focused analysis of Iron Ore, the most valuable export product within Subcategory 28: Metalliferous Ores. Although the primary scope is from 2015 to 2024, the timeline shown includes earlier years to highlight global shocks and trade agreements that shaped recent performance

Metalliferous Ores: Export Trends with Key Global and Trade Agreement Milestones

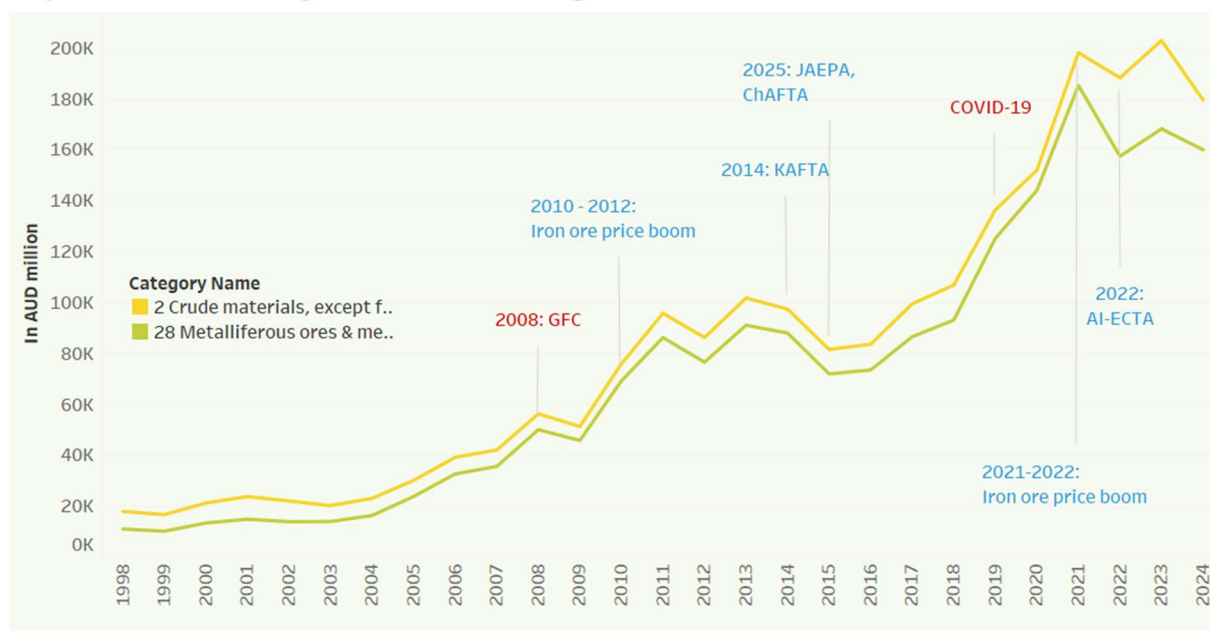


Figure 8.1: Iron Ore Export Trends with Global Events and FTA Milestones (1998-2024)

Between 2015 and 2024, iron ore exports experienced both volatility and growth. After a brief dip from 2015 to 2016, following the commodity price correction, the trend stabilised and gradually increased. This recovery was partly supported by ongoing demand from Asia and the foundation laid by earlier free trade agreements, particularly the Korea-Australia Free Trade Agreement (KAFTA) in 2014 and the ChAFTA (China-Australia FTA), which began implementation in 2015 (DFAT, 2023).

The most significant upturn occurred from 2020 to 2021 when iron ore exports surged despite the economic disruptions of COVID-19. This unexpected growth was driven by a sharp increase in iron ore prices, supported by supply constraints in Brazil and rising demand from China's stimulus-driven construction sector (Department of Industry, 2022). The chart shows a steep climb during this period, indicating a substantial gain in export value.

In 2022, the Australia-India Economic Cooperation and Trade Agreement (AI-ECTA) came into effect, contributing to continued export strength (DFAT, 2023). However, the growth pace slowed from 2023 to 2024, partly due to cooling demand and market adjustments post-pandemic. While values remained high, the rate of increase plateaued.

From 2015 to 2024, the net gain in iron ore export value is substantial. This period reflects the product's resilience to global shocks and the effectiveness of Australia's trade positioning through strategic FTAs. Iron ore continues to serve as a stabilising force in Australia's export economy, reinforcing its role as a long-term growth anchor.

8.1 5-Year Forecast (2025–2030)

The forecast model shown in *Figure 8.2* uses historical data from 1988 - 2024 to project export performance up to 2030. The central estimate suggests a compound annual growth rate (CAGR) of 7.8%, with export values expected to reach between AUD 190 billion and AUD 300 billion by 2030.

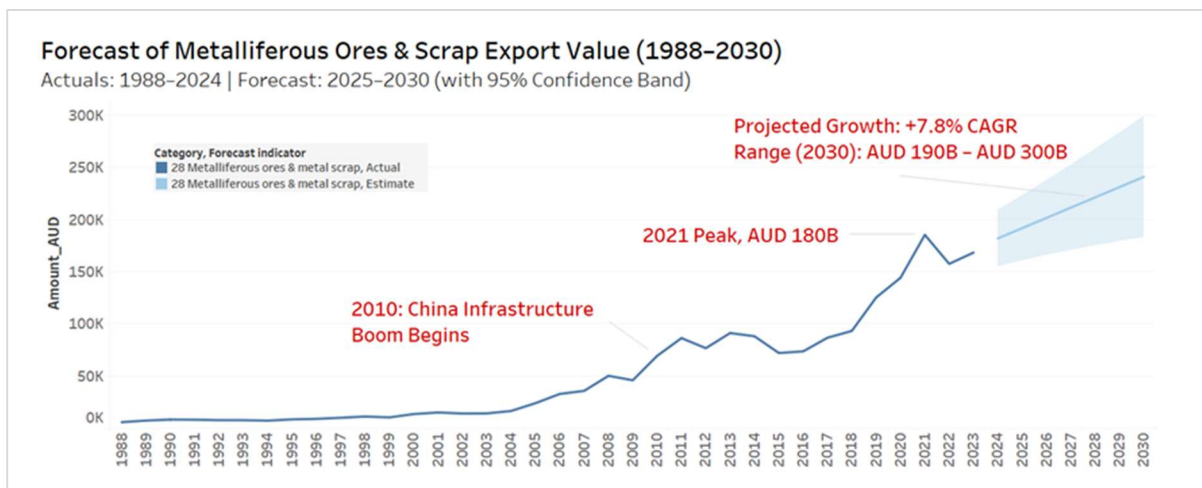


Figure 8.2: Iron Ore Export 5-year forecast trend (2025 - 2030)

The blue confidence band indicates increasing uncertainty in the forecast, reflecting potential external risks such as commodity price fluctuations, shifts in global demand, or geopolitical tensions. However, the upward trajectory suggests a strong long-term outlook, with sustained export growth expected under stable global conditions.

8.2 Recommendations

- **Strengthen Diplomatic and Trade Relations with Asia**
 - Given that China, Japan, South Korea, and India are the primary buyers, Australia should continue to prioritise and diversify trade agreements in Asia to secure long-term demand and reduce over-dependence on a single market.
- **Invest in Supply Chain Resilience and Logistics Efficiency**
 - Infrastructure improvements, particularly in rail and port capacity, should be prioritised to support growing export volumes, manage price volatility, and ensure efficient delivery and storage of ore.
- **Support Technological Innovation in Mining**
 - Encouraging advanced mining technologies can improve operational efficiency, reduce environmental impact, and enhance the industry's sustainability credentials, aligning with global ESG expectations.
- **Monitor Policy Shifts in Major Importing Countries**
 - Exporters and policymakers should closely watch regulations in key markets (e.g., China's steel decarbonisation or India's mining subsidies), which could affect demand and pricing.
- **Explore Value-Added Processing Opportunities**

- Rather than relying solely on raw ore exports, Australia can explore upstream opportunities in ore beneficiation and metal refining to increase value capture and create new industrial jobs.

9. Visualization Approach Summary

Visualization Approach	Advantages	Best Use Cases
Line Charts	Display trends and changes over time clearly.	Showing time-series patterns and event-driven trade movements.
Stacked Area Charts	Show part-to-whole relationships and how category shares evolve over time.	Visualising product composition and export/import structure shifts.
Treemap Charts	Show proportions compactly and highlight dominant elements visually.	Displaying country-level trade shares (export/import) within one glance.
Bar Charts (Horizontal & Diverging)	Compare absolute values or polarity (e.g., surplus vs deficit) across years.	Highlighting trade balance shifts or volatility by category.
Donut Charts	Show percentage composition of a category clearly.	Explaining component dominance (e.g., Iron Ore share within Subcategory 28).
Forecast Line Charts	Display projections with uncertainty range.	Communicating future trends with confidence levels.
Annotated Charts	Link data trends with key contextual events.	Enhancing narrative clarity in story-driven visualisations.
Composite Dashboards	Combine multiple chart types in one view for full-picture analysis.	Presenting a holistic overview of trends, forecasts, and composition efficiently.

10. Advantages and Disadvantages of the Dashboard and Storyboard Methodologies

The dashboard approach offered substantial advantages regarding interactivity and depth of analysis. It allowed users to explore long-term trends, identify category-level patterns, and drill down into Subcategory 28 for detailed insights. Multiple visual elements, such as trend lines, composition charts, and forecast models, helped present a comprehensive view of trade performance. However, the complexity of the dashboard may be challenging for first-time users, and the lack of a clear narrative structure can limit its effectiveness for non-technical audiences.

In contrast, the storyboard effectively communicated key findings through a structured, slide-by-slide narrative. Each visual focused on a specific aspect of the analysis, making it easier to highlight turning points, trends, and strategic insights. It was beneficial for presenting to stakeholders who require a clear and concise overview. Nonetheless, the static nature of storyboards limits interactivity, and viewers cannot explore beyond the predefined path.

Using both methods together balanced the need for deep exploration and clear communication. The dashboard supported data-driven discovery, while the storyboard translated those insights into a compelling story. This dual-method approach enhanced the overall analytical impact of the project.

11. Summary and Conclusion

This report analysed Australia's trade from 1988 to 2024, focusing on Crude Materials (SITC 2) and Subcategory 28: Metalliferous Ores. The study identified long-term export growth, consistent trade surpluses, and iron ore's strategic value using both dashboard and storyboard methodologies.

The dashboard provided interactive, multi-angle analysis, ideal for deeply exploring trends, volatility, and forecasts. However, it may overwhelm new users due to its complexity and lack of a linear storytelling structure.

The storyboard offered a straightforward, sequential narrative, compelling for communicating insights to broader audiences. It simplifies complex findings but is static and less flexible for exploration.

Combining both methods created a well-rounded analysis: dashboards allowed detailed investigation, while storyboards communicated key findings with clarity and impact.

The 5-year forecast suggests continued export growth for iron ore, supported by strong demand and infrastructure. Australia should focus on trade diversification, logistics investment, and mining innovation to maintain this edge.

Together, these insights reinforce the importance of Subcategory 28 not only as a driver of exports but as a pillar of Australia's long-term trade strategy.

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